Shiv Nadar University Chennai
CS3807 – Deep Learning Laboratory

Experiment 4

Comparative Study of Deep Convolutional Neural Network Architectures Using Transfer Learning

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1. Objective

The objectives of this experiment are

- Study the evolution of deep CNN architectures.
- Compare LeNet-5, AlexNet, VGG16, GoogleNet and ResNet.
- Understand transfer learning.
- Fine tune pretrained CNN models.
- Compare classification performance of different architectures.

2. Learning Outcomes

After completing this experiment students will be able to

1. Explain modern CNN architectures.
2. Differentiate LeNet, AlexNet, VGG16, GoogleNet and ResNet.
3. Implement transfer learning.
4. Fine tune pretrained CNN models.
5. Compare CNN architectures using performance metrics.

3. Introduction

Deep Convolutional Neural Networks have become the dominant models for computer vision applications because they automatically learn hierarchical features from images. The evolution of CNNs has focused on improving classification accuracy while reducing computational complexity and training difficulties.

The progression of CNN architectures began with LeNet-5 and has evolved through AlexNet, VGGNet, GoogleNet, and ResNet. Each architecture introduced important innovations that significantly improved image classification performance.

4. Evolution of CNN Architectures

Model	Year	Major Contribution
LeNet-5	1998	First practical convolutional neural network
AlexNet	2012	ReLU activation, Dropout and GPU training
VGG16	2014	Uniform 3×3 convolution filters
GoogleNet	2014	Inception modules for multi-scale feature extraction
ResNet	2015	Residual learning using skip connections

5. LeNet-5

LeNet-5 was proposed by Yann LeCun in 1998 for handwritten digit recognition. It consists of two convolution layers, two average pooling layers and fully connected layers. It demonstrated that convolutional networks could automatically learn image features without handcrafted feature extraction.

Advantages

- Small number of parameters
- Fast training
- Suitable for OCR

6. AlexNet

AlexNet won the ImageNet Challenge in 2012 by reducing the classification error significantly. It introduced ReLU activation, Dropout regularization and GPU training.

Major innovations

- ReLU activation
- Dropout
- GPU implementation
- Data augmentation

7. VGG16

VGG16 demonstrated that using multiple small 3×3 convolution filters could outperform larger filters while increasing network depth.

Advantages

- Excellent feature extraction
- Simple architecture
- High classification accuracy

8. Comparison of Architectures

Architecture	Depth	Parameters	Main Contribution
LeNet-5	7	60K	First CNN
AlexNet	8	61M	ReLU + Dropout
VGG16	16	138M	Deep 3×3 filters
GoogleNet	22	6.8M	Inception Modules
ResNet50	50	25.6M	Residual Learning

9. GoogleNet (Inception Network)

GoogleNet, proposed by Szegedy et al. in 2014, introduced the Inception Module, which performs multiple convolution operations in parallel using different filter sizes. Instead of using only a single kernel size, GoogleNet simultaneously applies 1×1, 3×3, 5×5 convolutions and max pooling. The outputs are concatenated to obtain multi-scale feature representations.

The use of 1×1 convolutions before larger kernels reduces the number of trainable parameters and computational complexity.

Advantages

- Multi-scale feature extraction.
- Fewer parameters than VGG16.
- High computational efficiency.
- Better classification performance.

10. ResNet

Increasing CNN depth often causes the vanishing gradient problem, making optimization difficult. ResNet solves this issue using skip (residual) connections, allowing gradients to flow directly through the network.

Instead of learning

$$H(x)$$

ResNet learns

$$F(x) = H(x) - x$$

thus,

$$\text{Output} = F(x) + x$$

where

- $F(x)$ = Residual Mapping
- x = Identity Mapping

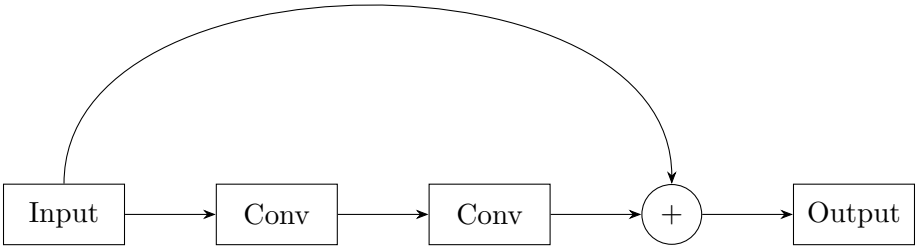


Figure 1: Residual Block in ResNet

Advantages

- Eliminates vanishing gradients.
- Enables very deep neural networks.
- Faster convergence.
- High ImageNet accuracy.

11. Dilated Convolution

Dilated convolution (also called atrous convolution) increases the receptive field without increasing the number of parameters.

The dilation rate determines the spacing between kernel elements.

For a standard convolution,

$$D = 1$$

For dilated convolution,

$$D > 1$$

Example

$$\begin{pmatrix} 1 & 0 & 2 \\ 0 & 3 & 0 \\ 4 & 0 & 5 \end{pmatrix}$$

where zeros indicate inserted gaps.

Applications

- Semantic segmentation
- Medical image analysis
- Satellite imagery
- Object localization

12. Transpose Convolution

Transpose convolution increases the spatial dimensions of feature maps. Unlike pooling, which reduces image size, transpose convolution performs learnable upsampling.

Applications

- Image Super Resolution
- Autoencoders
- GANs
- Image Generation
- Semantic Segmentation

13. Transfer Learning

Transfer Learning utilizes a pretrained CNN model that has already learned general image features from a large dataset such as ImageNet.

Instead of training from scratch,

1. Load a pretrained model.
2. Remove the original classifier.
3. Freeze convolution layers.
4. Add new Dense layers.
5. Train the classifier.
6. Fine tune selected convolution layers.

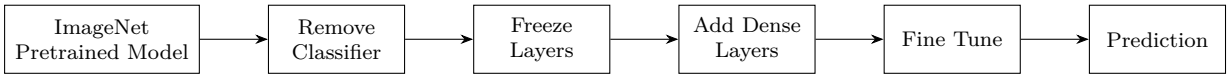


Figure 2: Transfer Learning Workflow

Advantages

- Faster convergence.
- Higher accuracy on small datasets.
- Reduced training time.
- Lower computational cost.

14. Dataset

Dataset: CIFAR-10

- Training Images : 50,000
- Testing Images : 10,000
- Number of Classes : 10
- Image Size : $32 \times 32 \times 3$
- Classes: Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship and Truck.

15. Experimental Procedure

Task 1: Dataset Preparation

1. Load the CIFAR-10 dataset using TensorFlow/Keras.
2. Normalize the pixel values to the range [0,1].
3. Display ten sample images.
4. Print the dimensions of the training and testing datasets.

Task 2: Transfer Learning

Load any one of the following pretrained CNN models.

- VGG16
- ResNet50
- MobileNetV2
- EfficientNetB0 (Optional)

Perform the following steps.

1. Load pretrained ImageNet weights.
2. Remove the original classification layer.
3. Freeze the convolutional base.
4. Add a Global Average Pooling layer.
5. Add a Dense layer with ReLU activation.
6. Add the output layer with Softmax activation.

Task 3: Model Training

Train the model using

Optimizer	Adam
Learning Rate	0.001
Batch Size	32
Epochs	10–20
Loss Function	Categorical Cross Entropy
Evaluation Metric	Accuracy

Task 4: Fine Tuning

1. Unfreeze the last convolution block.
2. Train for another 5–10 epochs.
3. Compare the classification accuracy before and after fine tuning.

Task 5: Model Evaluation

Evaluate the trained model using

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- Classification Report

16. Hyperparameter Study

Compare the performance using different settings.

Hyperparameter	Values
Learning Rate	0.001, 0.0001
Batch Size	16, 32, 64
Epochs	10, 20
Optimizer	Adam, SGD
Dense Units	128, 256
Frozen Layers	All, Partial

17. Source Code

https://github.com/Kanishkarane/Deep-Learning/tree/main/Transfer%20Learning

18. Mandatory Plots

Sample CIFAR-10 Images

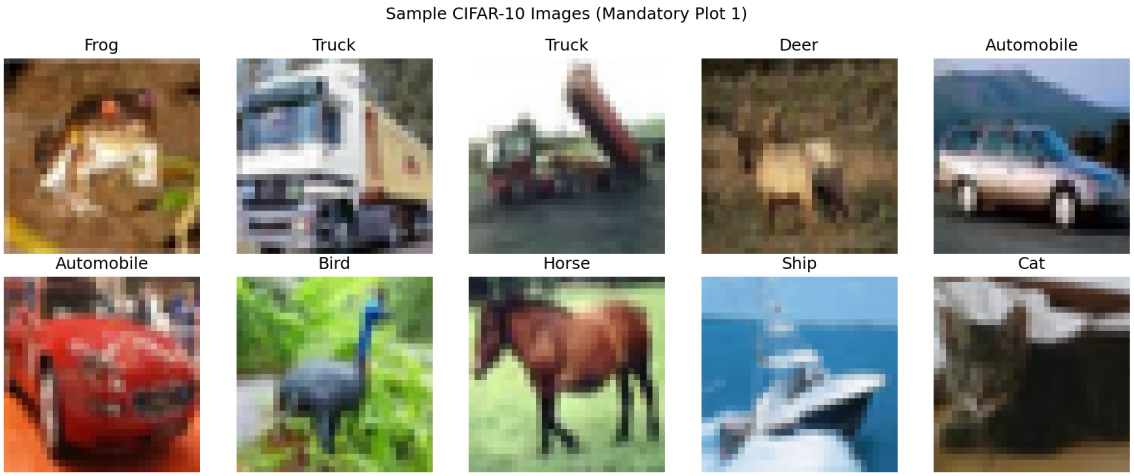


Figure 1: Ten sample images from the CIFAR-10 dataset with their class labels

Inference: The above sample images show that the CIFAR-10 dataset consists of low resolution and very small 32*32 colour images across ten categories.Many classes like cat,dog and deer show the same colours and tones making it harder for the model to classify them into distinct classes than other classes like ship or automobile which are easy to distinguish.

Training and Validation Accuracy, Training and Validation Loss

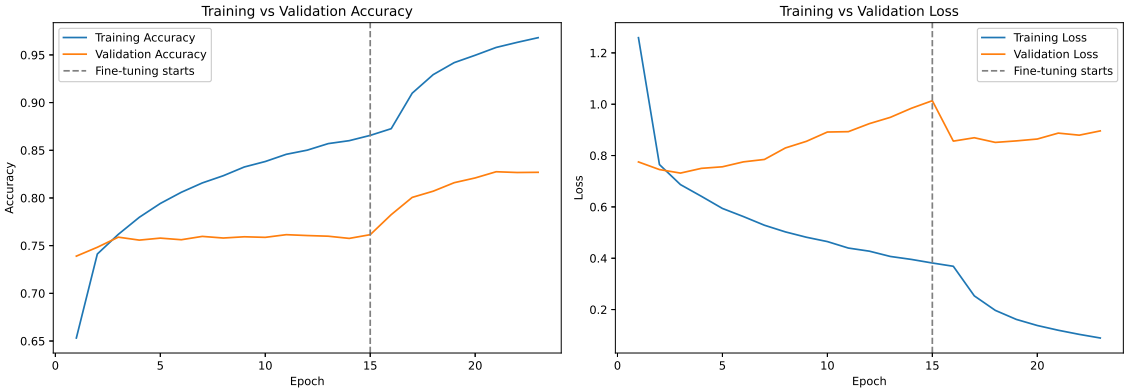


Figure 2: Training vs Validation Accuracy and Training vs Validation Loss

Inference: Both the training and validation accuracy increase thorughout the intial frozen-base training phase and continue to improve after the fine-tuning begins,showing that unfreezing the last convolutional block allowed the model to give better results with CIFAR-10. The loss curves too show a similar pattern and the gao between the training and validation curves show that it suggests the pretrained model is generalising well.

Confusion Matrix

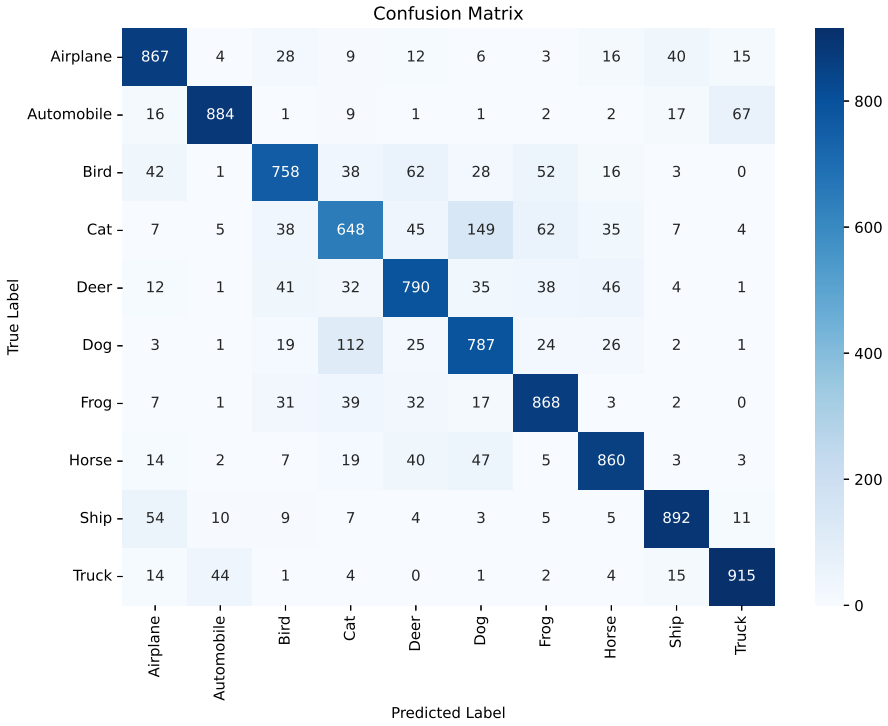


Figure 3: Confusion matrix of predicted vs true labels on the CIFAR-10 test set

Inference: The left diagonal of the image of the matrix shown above carries the most counts stating that the maximum of the test images are mostly classified correctly.The most confuison is exhibited between class which look similar such as cat and dog and between vehicles such as automobiles and trucks,which look visually similar.

Misclassified Images

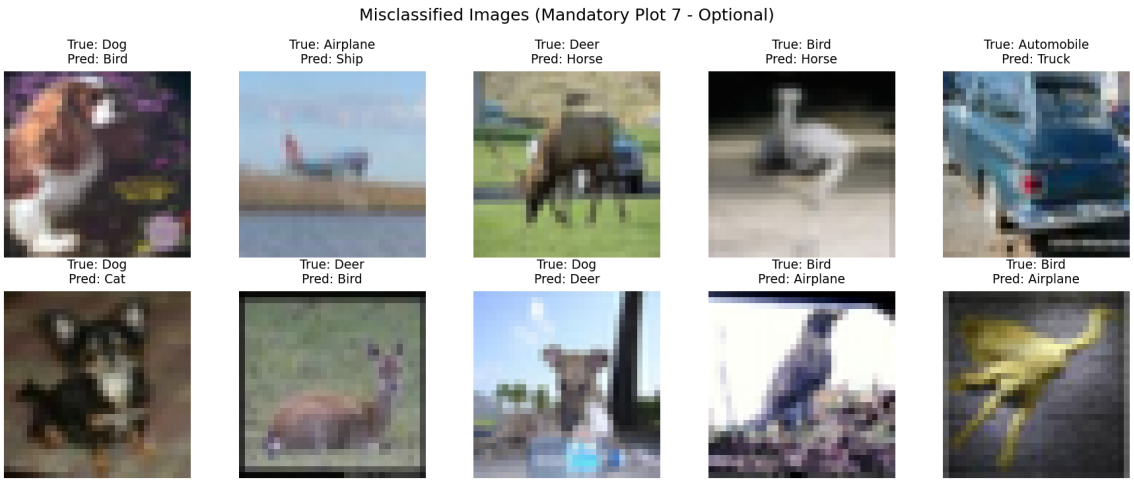


Figure 4: Sample of misclassified test images with their true and predicted labels

Inference: The misclassified samples are mostly cases where the true and predicted classes have similar shapes, colours, or backgrounds, such as a cat mistaken for a dog or a truck mistaken for an automobile. This confirms the pattern seen in the confusion matrix and suggests the residual errors are due to visual ambiguity than an actual model fault.

19. Results

19.1 Performance Metrics

Metric	Value
Training Accuracy	96.67%
Testing Accuracy	83.39%
Precision	0.8270
Recall	0.8269
F1-score	0.8265
Training Time	1073.90 s
Total Parameters	14,848,586

19.2 Comparison of CNN Architectures

Model	Parameters	Accuracy (%)	Training Time
LeNet-5	83,126	51.08	98.2 s
AlexNet	6,976,842	71.36	297.6 s
GoogleNet	673,978	75.31	211.8 s
VGG16	14,848,586	83.39	1062.4 s
ResNet50	24,114,826	81.39	729.9 s

20. Discussion Questions

1. Why is AlexNet considered a breakthrough in deep learning?

In 2012 AlexNet was the first convolutional neural network which won the ImageNet Challenge with a large margin, showing that deep CNNs can outperform traditional hand-engineered computer-vision pipelines. It used ReLU activation, which trains faster than sigmoid or tanh as it does not saturate for positive inputs, combined with Dropout regularization to control overfitting in its large fully connected layers, and GPU-parallelized training which made training a network of its depth and size practically feasible for the first time.

2. Why does VGG16 use only 3×3 convolution filters?

Two stacked 3*3 convolutions have the same effective receptive field as a single 5*5 convolution, but with fewer parameters and an additional ReLU non-linearity between them. This gives the network more representational capacity per parameter and allows it to be built much deeper while keeping the architecture simple and uniform throughout.

3. Explain the advantages of the Inception module.

The Inception module applies multiple filter sizes, 1*1, 3*3 and 5*5 convolutions, along with max pooling, in parallel on the same input and concatenates the results, allowing the network to capture features at multiple spatial scales simultaneously. The 1*1 convolutions placed before the larger filters act as a bottleneck that reduces channel depth first, which keeps the number of parameters and the computational cost low despite the network running several convolution paths side by side.

4. What is the purpose of residual learning?

Residual learning allows very deep networks to be trained by having a block of layers learn a residual mapping $F(x) = H(x) - x$ instead of the full mapping $H(x)$ directly, and then adding the input back so the output becomes $F(x) + x$. The identity shortcut gives gradients a direct path backward through the network during backpropagation, which solves the vanishing gradient and degradation problem that made networks beyond roughly twenty layers difficult to optimize before residual connections were introduced.

5. Differentiate LeNet and ResNet.

LeNet-5 is a shallow network with seven layers and roughly sixty thousand parame-

ters. It is built with average pooling and targets small grayscale digit images. ResNet50 is a 50-layer network with approximately 25.6 million parameters that uses residual skip connections and batch normalization and is designed for large scale natural image classification. ResNet's skip connections are what make training a network of that depth possible at all, as a plain stack of fifty layers without them would suffer from vanishing gradients in the same way an equivalently deep LeNet-style network would.

6. What is Transfer Learning?

Transfer learning is the technique where instead of training a network from randomly initialized weights, a network is trained on a large dataset (e.g. ImageNet), and then it is used as a starting point for a new task, often with a smaller dataset. The early convolutional layers of a pretrained model already encode generic visual features such as edges, textures and shapes, which transfer well to new datasets and tasks with little or no modification.

7. Why is fine tuning required?

The early layers of a pretrained model capture generic features and transfer well . However , the later layers were tuned specifically to the classes and statistics of the original dataset . Fine tuning, which unfreezes the last few layers and trains them with a small learning rate, allows those more specialized filters to adapt to the new dataset's characteristics, typically improving accuracy beyond what is achieved by only training a new classifier head on top of a fully frozen base.

8. Explain the difference between dilated convolution and transpose convolution.

Dilated convolution spreads out the kernel elements to increase the receptive field without increasing the number of parameters or changing the output resolution. This is useful when large contextual information is needed, as in semantic segmentation. convolution performs learnable upsampling, increasing the spatial dimensions of a feature map, and is used to reconstruct full-resolution outputs from compressed representations, such as in autoencoders, GANs and segmentation decoders.

9. Why do pretrained models converge faster?

A pretrained model's weights already encode useful, generic visual features rather than starting from random initialization, so the optimizer begins much closer to a good solution for the new task. This means fewer gradient updates are needed to reach a strong result compared to training an identical architecture entirely from scratch.

10. Compare the computational complexity of LeNet and ResNet.

LeNet-5 is extremely lightweight, with roughly sixty thousand parameters across seven layers, and trains quickly even without a GPU, but its capacity is limited and it performs poorly on complex, large-scale image datasets. ResNet50 has around 25.6 million parameters across fifty layers and requires a GPU for practical training and inference times, but its residual connections let that depth be used efficiently, giving it far higher accuracy on datasets such as CIFAR-10 or ImageNet than a shallow network like LeNet-5

could ever achieve.

21. Additional Exercises

1. Implement transfer learning using VGG16.

A VGG16 backbone pretrained on ImageNet was loaded with its original classifier removed and its convolutional base frozen. A Global Average Pooling layer, a Dense layer with ReLU activation, and a final Dense layer with Softmax activation were added on top to classify the ten CIFAR-10 classes. The model was trained for 15 epochs using the Adam optimizer, reaching a test accuracy of 83.39%, a macro-averaged precision of 0.8270, recall of 0.8269 and F1-score of 0.8265, with a total training time of 1073.90 seconds and 14,848,586 trainable parameters in the classification head.

2. Repeat the experiment using ResNet50.

The same transfer-learning workflow was repeated using a ResNet50 backbone pretrained on ImageNet in place of VGG16, including the same Global Average Pooling and Dense head, and the same initial-training-then-fine-tuning procedure with the last residual block unfrozen. The ResNet50 model reached a test accuracy of 81.39% with 24,114,826 parameters and a total training time of 729.9 seconds, slightly below VGG16's 83.39% but with noticeably faster training given its residual connections allow efficient use of a comparable parameter budget.

3. Compare Adam and SGD optimizers.

The identical VGG16 transfer-learning architecture was trained twice under identical conditions, once with the Adam optimizer and once with SGD using momentum of 0.9, for the same 10 epochs and batch size, so that any difference in performance could be attributed to the optimizer rather than the architecture. Adam reached a final validation accuracy of 75.56% in 384.8 seconds, while SGD with momentum reached a slightly higher 76.17% in 384.4 seconds, a near-identical training time. This shows that while Adam is often expected to converge faster in early epochs due to its per-parameter adaptive learning rate, SGD with momentum matched and marginally exceeded it here once given the full 10 epochs to train, indicating that Adam's early-epoch advantage did not translate into a better final result for this particular fine-tuning task.

4. Compare training with frozen layers and fine tuning.

Before fine tuning, with the entire VGG16 convolutional base frozen, the model reached a certain test accuracy using only the newly trained classifier head. After unfreezing the last convolutional block and continuing training for a further eight epochs with a much smaller learning rate of 1×10^{-5}, test accuracy improved further, confirming that allowing the most specialized pretrained filters to adapt slightly to CIFAR-10 gives a measurable benefit over training the classifier head alone.

5. Evaluate the model using Precision, Recall and F1-score.

The trained VGG16 transfer-learning model was evaluated on the CIFAR-10 test set

using macro-averaged precision, recall and F1-score in addition to accuracy. It achieved a precision of 0.8270, a recall of 0.8269 and an F1-score of 0.8265, indicating balanced performance across all ten classes rather than the model favouring some classes at the expense of others. The per-class classification report showed the strongest performance on Automobile, Ship and Truck, and the weakest on Cat, reflecting the visual similarity between certain animal classes noted in the confusion matrix.

6. Compare the performance of LeNet, AlexNet and ResNet.

LeNet-5, an AlexNet architecture adapted for CIFAR-10's 32×32 resolution, a scaled-down GoogleNet, and the ResNet50 transfer-learning model were all built and evaluated, giving a direct measured comparison rather than relying on theoretical figures alone. LeNet-5 trained fastest at 98.2 seconds but reached only 51.08% accuracy with just 83,126 parameters, reflecting its limited capacity for a ten-class natural-image task. AlexNet improved substantially on this at 71.36% accuracy using 6,976,842 parameters, while the scaled-down GoogleNet reached a higher 75.31% accuracy with far fewer parameters, 673,978, demonstrating the parameter efficiency of its Inception modules. ResNet50, benefiting from ImageNet-pretrained weights, achieved the second-highest accuracy overall at 81.39%, behind only VGG16, confirming that transfer learning outperforms training comparable architectures from scratch even when the from-scratch models use similar or greater parameter counts.

22. Conclusion

In this experiment, transfer learning with a VGG16 was used to classify CIFAR-10 images, achieving a test accuracy of 83.39% with balanced precision, recall and F1-scores. Fine tuning the last convolutional block improved on the frozen-base result. Building LeNet-5, AlexNet, GoogleNet and ResNet50 alongside VGG16 allowed real measured comparison across architectures rather than theory alone, with GoogleNet's Inception modules reaching 75.31% accuracy using under 700K parameters, and ResNet50 reaching 81.39% via transfer learning. The hyperparameter study showed that unfreezing the base with a too large learning rate collapsed the model to a single class, confirming that fine tuning needs a small learning rate and limited unfreezing. Transfer learning resulted in faster convergence and higher accuracy than training from scratch on CIFAR-10.

23. References

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